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Sub : Chemistry Assignment 5

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IUPAC

Chemistry

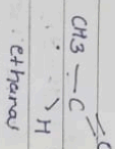
Assignment # 5

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Q1 The IUPAC name & structure of CH_3CHO is

- ethanal
- Acetaldehyde
- Acetone

answer (a) The IUPAC name of CH_3CHO is ethanal



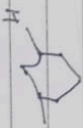
ethanal

Q2 The IUPAC name of $\text{CH}_3 - \text{O} - \text{C}_2\text{H}_5$ is

- methoxy ethane
- ethoxy ethane
- methyl ethane
- ethyl ethane

Ans. (a) The IUPAC name of $\text{CH}_3 - \text{O} - \text{C}_2\text{H}_5$ is methoxy ethane.

Q3 IUPAC name for



Ans. 3,3-dimethylcyclopentanol

Q4 Select the correct IUPAC name for

- 4-ethyl-3,3-dimethylhexane

4-ethyl-4,3-dimethylheptane

3-ethyl-3,4-dimethylheptane

3-ethyl-4,4-dimethylheptane

Ans. (a) 3-ethyl-4,4-dimethylheptane

Q5 The IUPAC structure of acetylsalicylic acid is

→ 2-acetoxy benzoic acid

Q6 Which are the names given to compound with double bonds.

Ans. Identify the smallest alkane which can form ring

Structure 1

→ (a) cyclopropane.

Q7 Select the correct IUPAC name of cyclopentane.

Ans. (a) the correct name of the above compound is 1,2-dimethylpropane.

Q8 Select the correct name of the following compound

Ans. (a) the correct name of the above compound is 1,2-dimethylpropane.

Q9 Identify the formula of ethanol.

Ans. (a) $\text{CH}_3\text{CH}_2\text{OH}$

Q.12. What is the order of priority of functional group?

Ans. The order of imp. of the functional group in the IUPAC system of nomenclature is: Carboxylic acid > Sulphonic acid > esters > acid halides > Amides > Cyanides > Aldehydes > Ketones > Alcohols > Amines > Alkenes > Alkynes > Alkyls.

Q.13. What is a functional group?

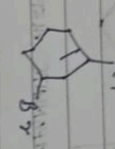
Ans. A functional group is the group of atoms in a molecule that determines the chemical behaviour of the molecule. It determines the chemical properties of the molecule.

Q.14. Give examples of a few functional groups and their formulae.

Ans. Examples of a few functional groups and their formulae are mentioned below:

Alcohol ($-OH$)	Sulphonic Acid ($-SO_3H$)
Aldehyde ($-CHO$)	Ketone ($-CO-$)
Ester ($-COOR$)	Cyanide ($-CN$)
Amide ($-CONH_2$)	Acid Halide ($-COCl$)

Q.15. The IUPAC name of compound.

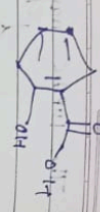


Ans. (a) The IUPAC name of the compound is 3-bromo-1-chlorocyclohexene.

Q.16. Match the following.

Q.16	Match the following
(i) Column 1	Column 2
(i) Salicylic Acid	1. $H-C(=O)-C(=O)-H$
(ii) Neopentane	2. $H-C(=O)-C(=O)-H$
(iii) o-Xylene	3. $H-C(=O)-C(=O)-H$
(iv) Toluene	4. $H-C(=O)-C(=O)-H$
(v) Acetylene	5. $CH_3-C \equiv C-CH_3$

Appendix: Directly related



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1. IUPAC NOMENCLATURE OF COMPOUNTS, WITH FUNCTIONAL GROUPS AND THE LONGEST CARBON

2. Find the longest carbon chain containing the functional group with the highest priority.

3. what is the IUPAC name of the following compound?

change the ending of the parent alkane alkene alkyne to the suffix of the highest priority group, which gives the parent name of the compound (usually drop the last letter 'e' before adding the suffix except for nitro where the 'e' is kept).

4. Number the chain from the end the closest to the highlighted functional group.

5. the other groups are named as substituents by using the appropriate prefix.

6. Assign stereochemistry, E/Z or R/S as necessary for naming purpose. The functional groups are assigned with priorities.

7. if the compound includes more than one functional group the one with the highest priority is the 'parent structure' and determines the 'parent name', the other groups will be regarded as substituents.

7. the "suffix" is used to indicate the name of the parent structure, and the prefix 's' is for the substituent.

8. The order of the group listed in tables is based on the decreasing order of priority, where the carbonyl group is in the highest priority.

9. The groups in the subordinate table have no difference in terms of priority, and they are usually listed in alphabetic order.

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